

Large power fluctuations in AI data centers - how serious is this problem?

Posted by u/brodds_c • 0 points • 5 comments

I'm conducting some research in this area. It seems like some software solutions have been implemented to address the large power fluctuations (e.g. scheduler, dummy load), but no good solution exists especially for the rapid load drops. On the hardware side, BESS designed for the grid seems way too expensive for load smoothing.

I read some reports from SemiAnalysis and Google, but it's not clear to me how difficult this problem is.

Is this something that can be effectively addressed with the currently available technology?

Comments

u/looktoward • 3 points • 2025-07-15 20:09 UTC

Some folks are using flywheels. Its a serious and difficult problem.

One solution - PyTorch PowerPlant No Blowup = 1

Which is a terrible solution.

u/brodds_c • 1 points • 2025-07-16 02:28 UTC

>PyTorch PowerPlant No Blowup = 1

How much wasted energy would result from this, percentage wise?

u/tokensRus • 2 points • 2025-07-15 08:28 UTC

Large Problem: [AI Training Load Fluctuations at Gigawatt-scale – Risk of Power Grid Blackout? – SemiAnalysis](<https://semianalysis.com/2025/06/25/ai-training-load-fluctuations-at-gigawatt-scale-risk-of-power-grid-blackout/>)

u/brodds_c • 1 points • 2025-07-16 01:57 UTC

yeah great piece by SemiAnalysis! Do you know how this problem is managed right now? How close are we to the point where data centers actually pose serious threat to the grid?

u/tokensRus • 2 points • 2025-07-16 05:53 UTC

Since i am a PR guy from Germany, i have not come to a solution yet. I have some journos wating for an article about this topic. What i know is that this is just the tip of the iceberg, since gpu clusters also tend to pulse, modern UPS systems can counter this effect...another solution should be the installation of a micro grid structure etc. but i am not an electrical engineer.